

Name: _____

Class: _____

Saturday, December 9, 2023

Test

Please use a pencil to complete the test. Print your name on the Name line above. Read the instructions for each section carefully. When you have completed the test, place your test face down on your desk and raise your hand.

- 1. In block codes, parity bits are _____ to block of messages.

1

a linear block code. In this code, a k -bit dataword is changed to an n -bit codeword where $n = k + 1$. The extra bit, called the *parity bit*, is selected to make the total number of 1s in the codeword even. Although some implementations specify an odd

- ☒ (A) Added ☐ (B) Subtracted
☐ (C) Multiplied ☐ (D) Divided

- 2. How many redundant bits are added in block codes for k information bits and n code bits?

1

ber of possible datawords. The block coding process is one-to-one; the same dataword is always encoded as the same codeword. This means that we have $2^n - 2^k$ codewords that are not used. We call these codewords invalid or illegal. The trick in

- ☐ (A) $n+k$ ☒ (B) $n-k$
☐ (C) k^2 ☐ (D) n^2

- 3. The ability of the block code to correct errors is a function of _____.

1

One of the central concepts in coding for error control is the idea of the Hamming distance. The **Hamming distance** between two words (of the same size) is the number of differences between the corresponding bits. We show the Hamming distance between two

- ☐ (A) Number of parity bits ☐ (B) Number of information bits
☐ (C) Number of code bits ☒ (D) Code distance

- 1 4. What is the relationship between the size of datawords (k) and the size of codewords (n) in block coding?

2^n codewords. Since $n > k$, the number of possible codewords is larger than the number of possible datawords. The block coding process is one-to-one; the same dataword is always encoded as the same codeword. This means that we have $2^n - 2^k$

- (A) $k = n$ (B) $n > k$
(C) $k > n$ (D) k is unrelated to n

- 1 5. If the receiver receives an invalid codeword in block coding, what does it suggest?

receiver receives an invalid codeword, this indicates that the data was corrupted during transmission.

- (A) The transmission was error-free (B) The data was not corrupted
(C) The data was corrupted during transmission (D) The codeword is valid but unused.

- 1 6. How can errors be detected using block coding?

How can errors be detected by using block coding? If the following two conditions are met, the receiver can detect a change in the original codeword.

1. The receiver has (or can find) a list of valid codewords.

- (A) By comparing the received codeword with the original dataword. (B) By checking if the received codeword matches any valid codeword.
(C) By verifying the length of the codeword. (D) By analyzing the parity of the original codeword.

- 1 7. In block coding for error detection, what happens if the received codeword is not valid?

word is accepted; the corresponding dataword is extracted for use. If the received codeword is not valid, it is discarded. However, if the codeword is corrupted during trans-

- (A) The codeword is accepted. (B) The dataword is extracted for use.
(C) The codeword is discarded. (D) The error remains undetected.

— 1 8. What does the Hamming distance measure between two words of the same size?

One of the central concepts in coding for error control is the idea of the Hamming distance. The **Hamming distance** between two words (of the same size) is the number of differences between the corresponding bits. We show the Hamming distance between two

- (A) Time elapsed during transmission (B) Number of differences between corresponding bits
(C) Signal strength during transmission (D) Length of the codewords

— 1 9. How can the Hamming distance be easily found between two words?

The Hamming distance can easily be found if we apply the XOR operation ($\oplus$) on the two words and count the number of 1s in the result. Note that the Hamming distance is a value greater than or equal to zero.

- (A) Apply the AND operation. (B) Apply the OR operation.
(C) Apply the XOR operation and count the number of 0s. (D) Apply the XOR operation and count the number of 1s.

— 1 10. What does the minimum Hamming distance represent in a set of codewords?

In a set of codewords, the **minimum Hamming distance** is the smallest Hamming distance between all possible pairs of codewords. Now let us find the minimum Hamming

- (A) Maximum distance between all codewords (B) Smallest distance between all possible pairs of codewords
(C) Average distance between all valid codewords (D) Total number of errors in codewords

— 1 11. What is the primary reason nonlinear block codes are not widely used for error detection and correction?

codes. The use of nonlinear block codes for error detection and correction is not as widespread because their structure makes theoretical analysis and implementation difficult. We therefore concentrate on linear block codes. The formal definition of linear

- (A) They are less efficient in terms of error correction. (B) Their structure makes theoretical analysis and implementation difficult.
(C) They are not capable of detecting errors. (D) Nonlinear block codes lack practical applications.

— 12. How is the parity bit calculated in the encoder of the parity-check code?

1

bit that is added makes the number of 1s in the codeword even. This is normally done by adding the 4 bits of the dataword (modulo-2); the result is the parity bit. In other words,

$$r_0 = a_3 + a_2 + a_1 + a_0 \quad (\text{modulo-2})$$

- (A) The sum of all bits in the dataword (B) The exclusive OR (XOR) of all bits in the dataword
- (C) The sum of the dataword bits using modulo-2 arithmetic (D) The difference between the largest and smallest bits in the dataword

— 13. What does the decision logic analyzer do when the syndrome is 0?

1

The syndrome is passed to the decision logic analyzer. If the syndrome is 0, there is no detectable error in the received codeword; the data portion of the received codeword is accepted as the dataword; if the syndrome is 1, the data portion of the received

- (A) Accepts the data portion of the received codeword as the dataword (B) Discards the data portion of the received codeword
- (C) Creates a new dataword (D) Increases the size of the codeword

— 14. How error detection and correction is done?

1

The central concept in detecting or correcting errors is redundancy.

- (A) By passing it through equalizer (B) By passing it through filter
- (C) By amplifying it (D) By adding redundancy bits

— 15. Which method requires less redundancy?

1

The correction of errors is more difficult than the detection. In **error detection**, we are only looking to see if any error has occurred. The answer is a simple yes or no. We are not even interested in the number of corrupted bits. A single-bit error is the same for us as a burst error. In **error correction**, we need to know the exact number of bits that are cor-

- (A) Error detection (B) Error correction
- (C) Error detection & correction (D) None of the mentioned

- 16. Which error detection method consists of just one redundant bit per data unit?
1

Parity-Check Code

Perhaps the most familiar error-detecting code is the **parity-check code**. This code is a linear block code. In this code, a k -bit dataword is changed to an n -bit codeword where $n = k + 1$. The extra bit, called the *parity bit*, is selected to make the total number of 1s in the codeword even. Although some implementations specify an odd number of 1s, we discuss the even case. The minimum Hamming distance for this category is $d_{\min} = 2$, which means that the code is a single-bit error-detecting code. Our first code (Table 10.1) is a parity-check code ($k = 2$ and $n = 3$). The code in Table 10.2 is also a parity-check code with $k = 4$ and $n = 5$.

- (A) Two-dimensional parity check (B) CRC
(C) Simple parity check (D) Checksum

- 17. Which of the following best describes a single-bit error?
1
from: https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

10.1.1 Types of Errors

Whenever bits flow from one point to another, they are subject to unpredictable changes because of **interference**. This interference can change the shape of the signal. The term **single-bit error** means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1. The term **burst error** means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1. Figure 10.1 shows the effect of a single-bit and a burst error on a data unit.

- (A) A) A single bit is inverted (B) B) A single bit per transmission is inverted
(C) C) A single bit per data unit is inverted (D) D) All of the choices are correct

- 18. How error detection and correction is done?
1 from: <https://www.sanfoundry.com/digital-communications-questions-answers-error-detection-and-correction-1/>

affect 10,000 bits.

10.1.2 Redundancy

The central concept in detecting or correcting errors is **redundancy**. To be able to detect or correct errors, we need to send some extra bits with our data. These redundant bits are added by the sender and removed by the receiver. Their presence allows the receiver to detect or correct corrupted bits.

- (A) a) By passing it through equalizer (B) b) By passing it through filter
(C) c) By amplifying it (D) d) By adding redundancy bits

- 19. Which error detection method involves polynomials?

1 Website:- Q(4)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

Note: cyclic codes called the cyclic redundancy check (CRC).

A better way to understand cyclic codes and how they can be analyzed is to represent them as polynomials. Again, this section is optional.

A pattern of 0s and 1s can be represented as a **polynomial** with coefficients of 0 and

- (A) a) CRC (B) b) Simple parity check
(C) c) Two-dimensional parity check (D) d) Checksum

- 20.
1 In cyclic redundancy checking, what forms the check bits?

Website: Q(7)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

When there are no bits left to pull down, we have a result. The 3-bit remainder forms the **check bits** (r_2 , r_1 , and r_0). They are appended to the dataword to create the codeword.

- (A) d) The dividend (B) a) The remainder
(C) c) The quotient (D) b) The divisor

— 21.

1

In CRC, if the dataword is 111111, the divisor 1010, and the remainder 110, what is the codeword at the receiver?

Website: Q(8)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

the remainder ($r_2r_1r_0$) is appended to the dataword to create the codeword.

The decoder receives the codeword (possibly corrupted in transition). A

(A) a) 111111011

(B) b) 1010110

(C) c) 111111110

(D) d) 110111111

- 22. In CRC, if the dataword is 111111 and the divisor 1010, what is the dividend at the sender?
1

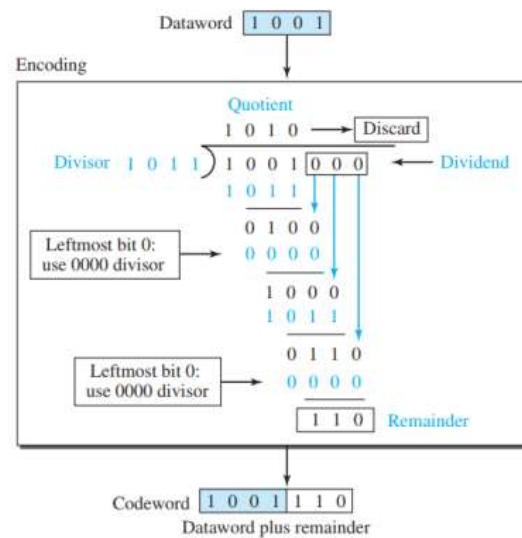
Website: Q(9)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

Encoder

Let us take a closer look at the encoder. The encoder takes a dataword and augments it with $n - k$ number of 0s. It then divides the augmented dataword by the divisor, as shown in Figure 10.6.

Figure 10.6 Division in CRC encoder



Note:

Multiply: AND
Subtract: XOR

- (A) a) 1111110000 (B) b) 111111000
(C) c) 111111 (D) d) 1111111010

— 23.

1

At the CRC generator, _____ is (are) added to the dataword after the division process to create the codeword.

Website: Q(10)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

$n - k + 1$ (4 here), predefined and agreed upon. The generator divides the augmented dataword by the divisor (modulo-2 division). The quotient of the division is discarded; the remainder ($r_2r_1r_0$) is appended to the dataword to create the codeword.

- ☐ (A) a) 0's ☐ (B) b) 1's
☒ (C) c) the remainder ☐ (D) d) the divisor

— 24.

1

In CRC, the quotient at the sender _____.

Website: Q(12)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

$n - k + 1$ (4 here), predefined and agreed upon. The generator divides the augmented dataword by the divisor (modulo-2 division). The quotient of the division is discarded; the remainder ($r_2r_1r_0$) is appended to the dataword to create the codeword.

- ☐ (A) a) becomes the dividend at the receiver ☐ (B) b) becomes the divisor at the receiver
☐ (C) c) is the remainder ☒ (D) d) is discarded

— 25. At the CRC checker, _____ means that the dataword is damaged.

1

Website: Q(13)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

process as the encoder. The remainder of the division is the syndrome. If the syndrome is all 0s, there is no error with a high probability; the dataword is separated from the received codeword and accepted. Otherwise, everything is discarded.

- ☐ (A) a) a string of alternating 1s and 0s ☒ (B) b) a nonzero remainder
☐ (C) c) a string of 0s ☐ (D) d) None of the choices are correct

— 26.

1

A codeword of 10 bits has only four 0s, how many terms are in the polynomial representation of this code?

Webite: Q(14)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

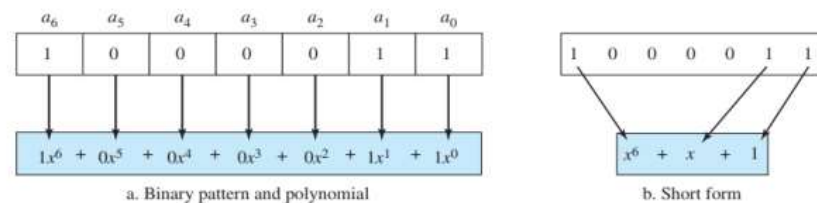
them as polynomials. Again, this section is optional.

A pattern of 0s and 1s can be represented as a **polynomial** with coefficients of 0 and 1. The power of each term shows the position of the bit; the coefficient shows the value of the bit. Figure 10.8 shows a binary pattern and its polynomial representation. In Figure 10.8a we show how to translate a binary pattern into a polynomial; in Figure 10.8b we show how the polynomial can be shortened by removing all terms with zero coefficients and replacing x^1 by x and x^0 by 1.

Figure 10.8 shows one immediate benefit; a 7-bit pattern can be replaced by three terms. The benefit is even more conspicuous when we have a polynomial such as

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Figure 10.8 A polynomial to represent a binary word



$x^{23} + x^3 + 1$. Here the bit pattern is 24 bits in length (three 1s and twenty-one 0s) while the polynomial is just three terms.

(A) 4

(C) 8

(B) 6

(D) d) None of the choices are correct

— 27.

1

In CRC, if the remainder is only three bits, the divisor should be _____ bits.

Website: Q(15)

https://highered.mheducation.com/sites/0073376221/student_view0/chapter10/quizzes.html

of the word. The n -bit result is fed into the generator. The generator uses a divisor of size $n - k + 1$ (4 here), predefined and agreed upon. The generator divides the augmented dataword by the divisor (modulo-2 division). The quotient of the division is discarded; the remainder ($r_2r_1r_0$) is appended to the dataword to create the codeword.

The decoder receives the codeword (possibly corrupted in transition). A copy of all n bits is fed to the checker, which is a replica of the generator. The remainder produced

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by the checker is a syndrome of $n - k$ (3 here) bits, which is fed to the decision logic

- | | |
|---|--|
| ☐ (A) a) 3 | ☐ (B) b) 2 |
| ☒ (C) c) 4 | ☐ (D) d) None of the choices are correct |

— 28. How does CRC work?

1

The process of modulo-2 binary division is the same as the familiar division process we use for decimal numbers. However, addition and subtraction in this case are the same; we use the XOR operation to do both.

- | | |
|--|---|
| ☐ (A) a) It adds redundant bits to the data for error correction | ☒ (B) b) It performs bitwise XOR operations on the data |
| ☐ (C) c) It encrypts the data using a secret key | ☐ (D) d) It compresses the data using lossless compression algorithms |

— 29. What is the role of the syndrome in the CRC decoder?

1

process as the encoder. The remainder of the division is the syndrome. If the syndrome is all 0s, there is no error with a high probability; the dataword is separated from the received codeword and accepted. Otherwise, everything is discarded.

- | | |
|--|---|
| ☐ (A) a) It is used to encrypt the received codeword. | ☐ (B) b) It indicates the type of error correction needed. |
| ☐ (C) c) It is the remainder obtained from dividing the received codeword. | ☒ (D) d) It helps decide whether there is an error in the received codeword. |

- 30. How does the CRC decoder determine whether there is an error in the received
1 codeword?

process as the encoder. The remainder of the division is the syndrome. If the syndrome is all 0s, there is no error with a high probability; the dataword is separated from the received codeword and accepted. Otherwise, everything is discarded.

- (A) a) By comparing the received codeword with the original dataword. (B) b) By analyzing the parity bits in the received codeword.
- (C) c) By dividing the received codeword by the generator polynomial. (D) d) By checking if the syndrome is all 0s.

- 31. What is the significance of using XOR (exclusive OR) operation in the modulo-2 binary
1 division process, as described in the encoder's operation?

The process of modulo-2 binary division is the same as the familiar division process we use for decimal numbers. However, addition and subtraction in this case are the same; we use the XOR operation to do both.

As in decimal division, the process is done step by step. In each step, a copy of the divisor is XORed with the 4 bits of the dividend. The result of the XOR operation (remainder) is 3 bits (in this case), which is used for the next step after 1 extra bit is pulled down to make it 4 bits long. There is one important point we need to remember in this type of division. If the leftmost bit of the dividend (or the part used in each step) is 0, the step cannot use the regular divisor; we need to use an all-0s divisor.

When there are no bits left to pull down, we have a result. The 3-bit remainder forms the **check bits** (r_2 , r_1 , and r_0). They are appended to the dataword to create the codeword.

- (A) a) XOR ensures encryption of the dataword. (B) b) XOR is a substitute for addition in binary division.
- (C) c) XOR guarantees error correction in the codeword. (D) d) XOR enhances data compression.

- 32. What is the significance of representing a binary pattern as a polynomial in the context of cyclic codes?

10.3.2 Polynomials

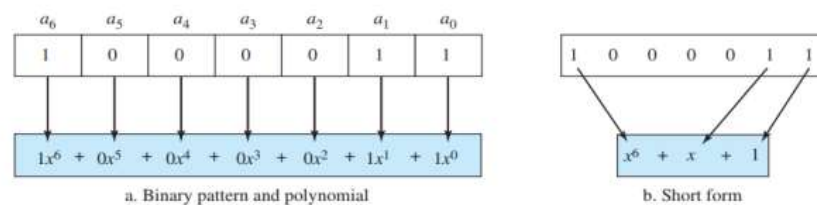
A better way to understand cyclic codes and how they can be analyzed is to represent them as polynomials. Again, this section is optional.

A pattern of 0s and 1s can be represented as a **polynomial** with coefficients of 0 and 1. The power of each term shows the position of the bit; the coefficient shows the value of the bit. Figure 10.8 shows a binary pattern and its polynomial representation. In Figure 10.8a we show how to translate a binary pattern into a polynomial; in Figure 10.8b we show how the polynomial can be shortened by removing all terms with zero coefficients and replacing x^1 by x and x^0 by 1.

Figure 10.8 shows one immediate benefit; a 7-bit pattern can be replaced by three terms. The benefit is even more conspicuous when we have a polynomial such as

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Figure 10.8 A polynomial to represent a binary word



- (A) a) It allows for the encryption of the binary pattern. (B) b) It simplifies the representation of the pattern using mathematical operations.
- (C) c) It increases the length of the binary pattern. (D) d) It introduces a random factor in the cyclic code.

- 33. How is the degree of a polynomial related to the number of bits in the corresponding
1 binary pattern, according to the information provided in the passage?

Degree of a Polynomial

The degree of a polynomial is the highest power in the polynomial. For example, the degree of the polynomial $x^6 + x + 1$ is 6. Note that the degree of a polynomial is 1 less than the number of bits in the pattern. The bit pattern in this case has 7 bits.

- (A) a) The degree of a polynomial is equal to the number of bits in the pattern.
- (B) b) The degree of a polynomial is one less than the number of bits in the pattern.
- (C) c) The degree of a polynomial is one more than the number of bits in the pattern.
- (D) d) The degree of a polynomial is twice the number of bits in the pattern.
- 34. How are addition and subtraction of polynomials performed in modulo-2 arithmetic,
1 and what are the consequences of this arithmetic in terms of combining terms?
- a) Addition and subtraction are different operations, and combining terms is not allowed.
- b) Addition and subtraction are the same operation, and combining terms involves deleting pairs of identical terms.
- c) Addition and subtraction are performed using multiplication.
- d) Combining terms is not applicable in modulo-2 arithmetic.

Adding and Subtracting Polynomials

Adding and subtracting polynomials in mathematics are done by adding or subtracting the coefficients of terms with the same power. In our case, the coefficients are only 0 and 1, and adding is in modulo-2. This has two consequences. First, addition and subtraction are the same. Second, adding or subtracting is done by combining terms and deleting pairs of identical terms. For example, adding $x^5 + x^4 + x^2$ and $x^6 + x^4 + x^2$ gives just $x^6 + x^5$. The terms x^4 and x^2 are deleted. However, note that if we add, for example, three polynomials and we get x^2 three times, we delete a pair of them and keep the third.

- (A) a) Addition and subtraction are different operations, and combining terms is not allowed.
- (B) b) Addition and subtraction are the same operation, and combining terms involves deleting pairs of identical terms.
- (C) c) Addition and subtraction are performed using multiplication.
- (D) d) Combining terms is not applicable in modulo-2 arithmetic.

- 35. What is the purpose of shifting in polynomial arithmetic, and how is it achieved in
1 both left and right directions?

b) Shifting is done for compression, and it is achieved by dividing each term by a fixed factor.

Shifting

A binary pattern is often shifted a number of bits to the right or left. **Shifting to the left means adding extra 0s as rightmost bits; shifting to the right means deleting some rightmost bits.** Shifting to the left is accomplished by multiplying each term of the polynomial by x^m , where m is the number of shifted bits; shifting to the right is accomplished by dividing each term of the polynomial by x^m . The following shows shifting to the left and to the right. Note that we do not have negative powers in the polynomial representation.

Shifting left 3 bits: 10011 becomes 10011000

$x^4 + x + 1$ becomes $x^7 + x^4 + x^3$

Shifting right 3 bits: 10011 becomes 10

$x^4 + x + 1$ becomes x

- (A) a) Shifting is done for encryption, and it is achieved by multiplying each term by a random factor. (B) b) Shifting is done for compression, and it is achieved by dividing each term by a fixed factor.
- (C) c) Shifting is done to the left by adding extra 0s, and to the right by deleting rightmost bits. (D) d) Shifting is done for error correction, and it is achieved by adding extra parity bits.

- 36. What type of codes are cyclic codes?
1

Cyclic codes are special linear block codes with one extra property. In a **cyclic code**, if a codeword is cyclically shifted (rotated), the result is another codeword. For example, if 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword. In this case, if we call the bits in the first word a_0 to a_6 , and the bits in the second word b_0 to b_6 , we can shift the bits by using the following:

$$b_1 = a_0 \quad b_2 = a_1 \quad b_3 = a_2 \quad b_4 = a_3 \quad b_5 = a_4 \quad b_6 = a_5 \quad b_0 = a_6$$

In the rightmost equation, the last bit of the first word is wrapped around and becomes the first bit of the second word.

- (A) a) Linear block codes (B) b) Non-linear codes
- (C) c) Convolutional codes (D) d) Hamming codes

— 37. How is division performed in polynomial representation?

1

of the quotient is then x^6/x^3 , or x^3 . Then we multiply x^3 by the divisor and subtract (according to our previous definition of subtraction) the result from the dividend. The

- (A) By multiplying the dividend and divisor
- (B) By subtracting the divisor from the dividend
- (C) By adding the divisor to the dividend
- (D) By dividing the coefficients of the dividend by the coefficients of the divisor

— 38. How is the dataword "1001" represented in polynomial form?

1

word from a dataword. Figure 10.9 is the polynomial version of Figure 10.6. We can see that the process is shorter. The dataword 1001 is represented as $x^3 + 1$. The divisor 1011

- (A) $x^3 + x + 1$
- (B) $x^3 + 1$
- (C) $x^4 + x$
- (D) $x^2 + x + 1$